

A Minor Project II Proposal on

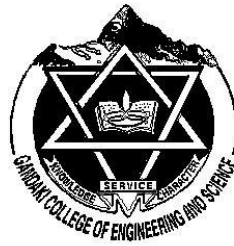
NeoNote

Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering in Software Engineering at Pokhara University

By

Anil B.K

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Department of Research and Development

GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Lamachaur, Kaski, Nepal

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APPROVAL CERTIFICATE

This project entitled "**NeoNote**" prepared and submitted by "**Anil BK**" and "**Kriti Adhikari**" under the supervision of "**Supervisor Name**" in partial fulfillment of the requirements for the Degree of Bachelor of Engineering in Software Engineering has been examined and is recommended for approval and acceptance.

Date of Evaluation: December 19, 2024

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(Project Supervisor)

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Er. Santosh Panth

Project Head

Research Management Committee

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ABSTRACT

NeoNote is a comprehensive independent app developed to address the need for efficient task management, note-taking, and project organization. The project aims to simplify the complexities of everyday planning by providing a unified platform that combines intuitive note-taking capabilities with robust task-tracking tools. Built using Flutter for the frontend, Django for the backend, and MySQL for database management, the system ensures seamless functionality, scalability, and secure data handling. The methodology involves creating an interactive interface that enables users to create, organize, and edit notes while managing tasks through features like customizable templates, real-time collaboration, and multi-device synchronization. Initial testing highlights smooth user interactions, reliable performance, and a responsive design. The system's effectiveness is evaluated through internal performance metrics and user trials, ensuring its suitability for both individual and collaborative use cases. **NeoNote** concludes by offering a versatile, efficient, and user-centric solution that empowers individuals and teams to optimize their productivity and achieve their goals.

Keywords: Independent app, task management, note-taking, project organization, productivity, Flutter, Django, MySQL, multi-device synchronization, real-time collaboration, scalable, secure data handling, user-centric, performance optimization, customizable templates.

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Chapter 1

INTRODUCTION

1.1 Background

NeoNote is an independent app designed to streamline task management, note-taking, and project organization. The primary objective of this platform is to simplify and enhance users' productivity by providing an intuitive and feature-rich tool that integrates all essential productivity features in one place. **NeoNote** offers a flexible and user-friendly approach, allowing users to create, organize, and track their tasks and notes easily. It includes customizable templates, real-time collaboration, and synchronization across devices for seamless accessibility. The app's modular design makes it adaptable to users' unique needs, enabling them to personalize their workspace. NeoNote allows users to visually track task progress, helping them stay on top of their goals and effectively manage their time. The system also provides users with the flexibility to categorize and organize their notes and tasks in a way that suits their workflow. This comprehensive tool ensures users can maintain clarity, stay focused, and work more efficiently toward their objectives.

1.2 Problem statement

- **Data Synchronization:** Ensuring real-time sync across devices without conflicts.
- **Scalability:** Maintaining performance as the app grows with more users and data.
- **Data Security:** Protecting user data from unauthorized access.
- **Collaboration Issues:** Handling real-time editing and user permissions effectively.

1.3 OBJECTIVE

This project will serve the following objectives:

- **Seamless Note-Taking:** Provide an intuitive platform for creating, editing, and organizing notes, tasks, and projects efficiently.
- **Cross-Platform Sync:** Sync data across desktop, laptop, and mobile devices for anytime, anywhere access.
- **Real-Time Collaboration:** Enable shared access to notes, tasks, and workspaces for team collaboration.
- **Customizable Templates:** Offer pre-built and editable templates for simplified organization.
- **Task Management:** Set deadlines, track progress, and manage tasks within a single interface.
- **Secure Data Storage:** Ensure reliable and secure storage of user data.
- **User-Friendly Design:** Create a clean and intuitive interface for seamless user experience.
- **Boost Productivity:** Provide tools to enhance organization and help users achieve goals efficiently.

1.4 IMPLICATION

NeoNote simplifies task management, note-taking, and project organization, improving productivity and teamwork. Its features, like cross-device access and real-time collaboration, help users stay organized and focused. These projects empower individuals and teams to work smarter, save time, and achieve their goals efficiently.

Chapter 2

LITERATURE REVIEW

Several products like **Trello**, **Google Keep**, **ClickUp**, and **Notion** offer similar features, such as task management and note-taking. However, **NeoNote** uniquely combines these features with real-time collaboration, customizable templates, and multi-device sync, all in an intuitive, user-friendly platform. Unlike others, it is specifically designed to streamline both personal and collaborative project management.

Table 1: COMPARISON TABLE

Application Features	Trello	Notion	Google Keep	ClickUp	NeoNote
Ease of use	MEDIUM	MEDIUM	HIGH	MEDIUM	HIGH
Task Management	HIGH	HIGH	MEDIUM	HIGH	HIGH
Note-Taking	LOW	HIGH	HIGH	MEDIUM	HIGH
Password Protection	YES	YES	YES	YES	YES
Team Management	YES	YES	NO	YES	YES
Real-Time Collaboration	YES	YES	YES	YES	YES

Chapter 3

TOOLS AND METHODOLOGY

3.1 REQUIRED TOOLS

Different tools used to develop our app:

- **Flutter:** Best for cross-platform deployment and creating visually rich, interactive apps.
- **Django:** For backend development, handling server-side logic and database management.
- **MySQL/PostgreSQL:** For managing the system's relational database, storing user data, notes, and tasks.
- **Firebase Authentication:** For handling user authentication and secure login/signup processes.
- **Firebase Real time Database:** For real-time data synchronization, enabling collaboration between users.
- **MS-Word:** For writing and preparing reports.

3.2 METHODOLOGY

For the development of **NeoNote**, an **Agile methodology** will be used to ensure flexibility and continuous improvement throughout the project. This iterative approach allows for frequent updates, quick feedback loops, and the ability to adapt to changing user needs. Each sprint will focus on specific features, such as task management, note-taking, and real-time collaboration, ensuring that the app evolves efficiently while maintaining a user-centric focus. This methodology also supports frequent testing and refinement to guarantee optimal performance and user satisfaction.

3.3 TESTING PLANS

Software testing is a critical element of software quality assurance and represents the ultimate review of specification, design and code generation. Software testing is an activity to check whether the actual results match the expected results and to ensure that the software system is defect free. It involves execution of a software component or system component to evaluate one or more properties of interest.

3.3.1 OBJECTIVE OF TESTING

- The first and the foremost objective of the software testing are to ensure that it fulfills all the requirements of the user, which means firstly understanding the user requirements and then ensuring it behaves in the similar manner.
- To providing a better user experience.
- To evaluate security related issues for the website under test.
- To evaluate the overall performance of the website.
- To uncover an as-yet-discovered error.

3.3.2 TEST CASES

Table 2: TEST CASE

Test Case ID	Purpose	Test Case	Expected Result
TC1	Authentication	Enter empty id and password	The id and password must be empty.
TC2	Authentication	Enter incorrect password	Password must be incorrect.
TC3	Check Invalid Password	Set password less than 8 character	The password must at least 10 character.
TC4	Authentication	Enter delete id and	This id shouldn't

		password of left user	exist.
TC5	Ensure that users can create a new note successfully.	Create a New Note	The new note is saved and visible in the note list.
TC6	Ensure changes made by one user are reflected instantly for another user.	Real-Time Collaboration (Note Editing)	Users see the changes immediately.
TC7	Verify that users can create, edit, and delete tasks.	Task Management	The task is created, edited, and deleted successfully, with changes reflected in the app.

3.4 Use-Case Diagram

USE CASE: NeoNote

Actors:

1. **User:** A person who uses NeoNote for note-taking, task management, and project organization.
2. **Admin:** Responsible for system maintenance, user management, and monitoring real-time collaboration.

Use Cases:

1. **Login:** Users and administrators can log in to the system using Firebase Authentication.
2. **Logout:** Users and administrators can log out of the system at any time.
3. **Create Note:** Users can create, edit, and save new notes.
4. **Organize Tasks:** Users can add, edit, and delete tasks related to their notes and projects.
5. **Real-Time Collaboration:** Users can collaborate on shared notes and see changes made by other users instantly.
6. **View Notes/Tasks:** Users can view their saved notes and tasks, organized by projects or categories.
7. **Set Reminders:** Users can set reminders for tasks or notes to receive notifications at a specified time.
8. **Share Notes:** Users can share notes with other users for collaborative editing and viewing.
9. **Sync across Devices:** Users can access their notes and tasks from multiple devices, with all data synchronized in real time.
10. **Delete Note/Task:** Users can delete any note or task they no longer need.

3.5 Entity Relation Diagram (EDR)

3.6 System Sequence Diagram (SSD

CHAPTER 4

TIME LINE CHART

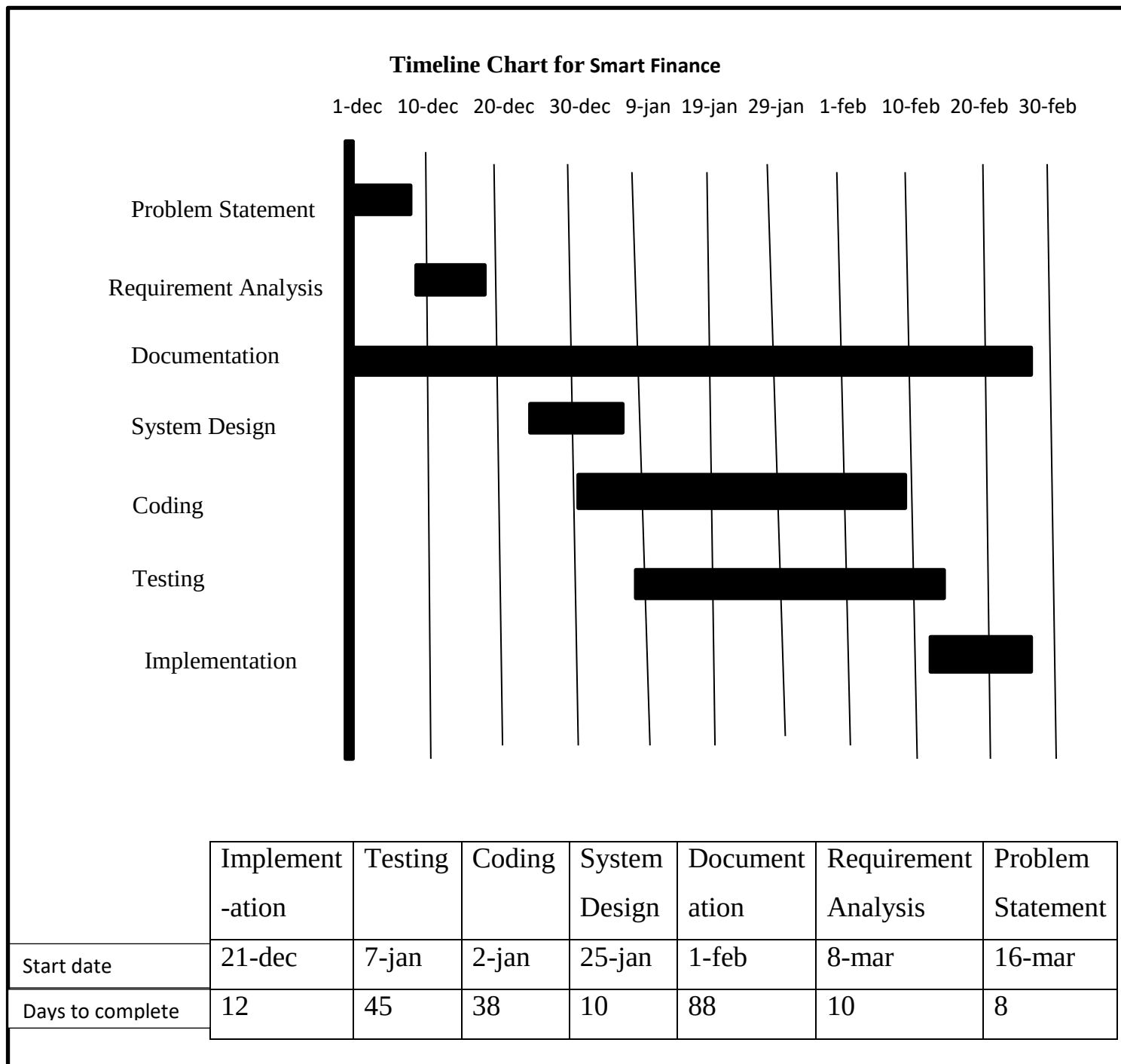


Fig 4: Timeline Chart

CHAPTER 5

EXPECTED OUTCOMES

After analyzing the requirements and defining the goals for **NeoNote**, we anticipate the following outcomes:

- Users can easily create, organize, and manage notes and tasks.
- Real-time collaboration allows changes to be instantly reflected for all users.
- Notes and tasks are synchronized across multiple devices.
- The app enhances productivity with customizable templates, reminders, and task tracking.
- A user-friendly interface ensures an intuitive experience.
- The system provides scalability and secure data management.

5.1 WIREFRAME

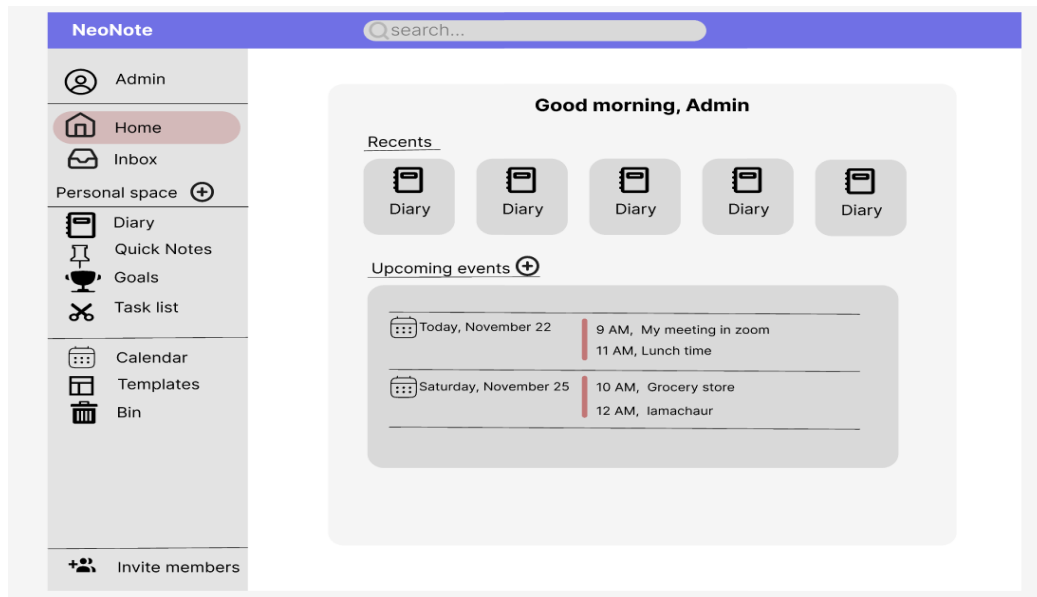


Fig 5.1 Home page 1

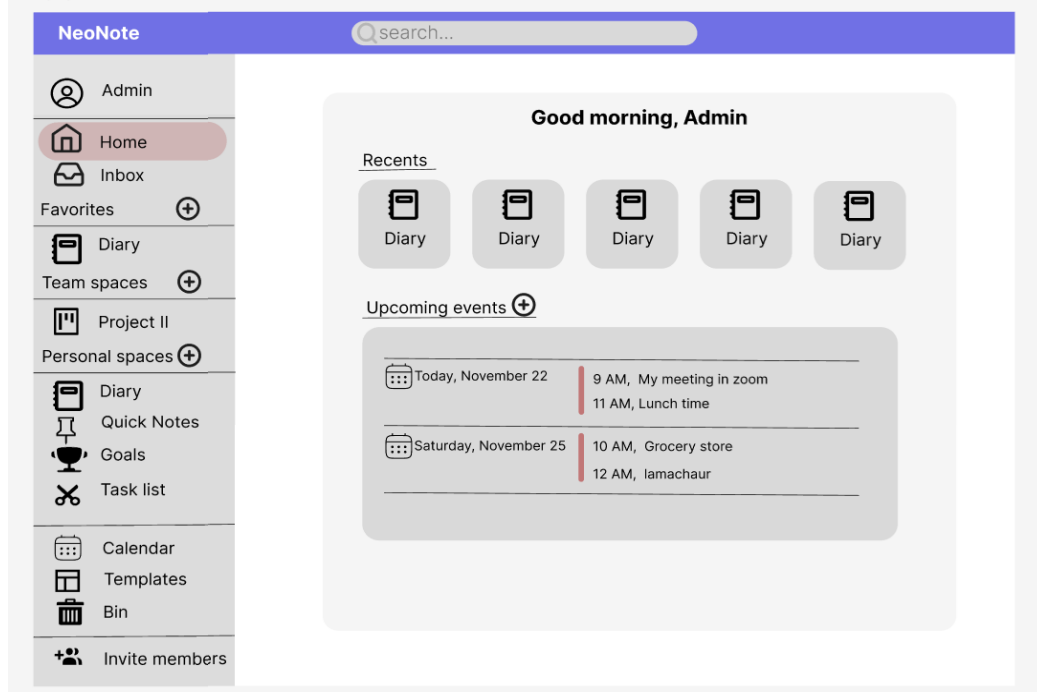


Fig 5.2 Home page 2

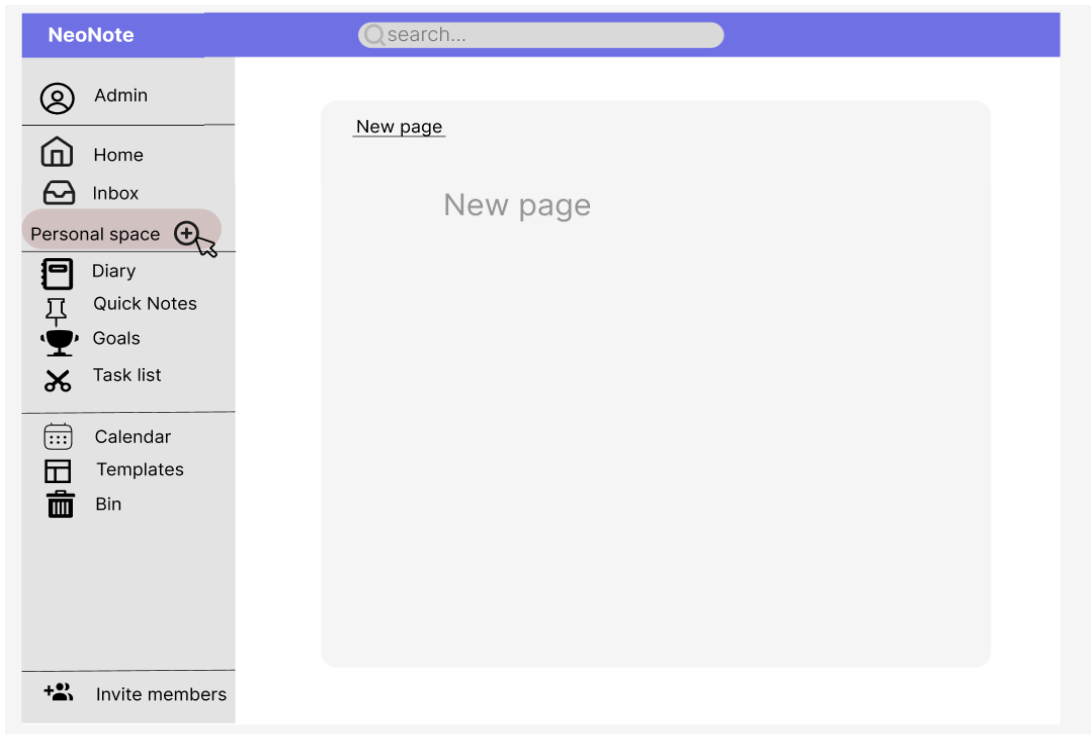


Fig 5.3 New page

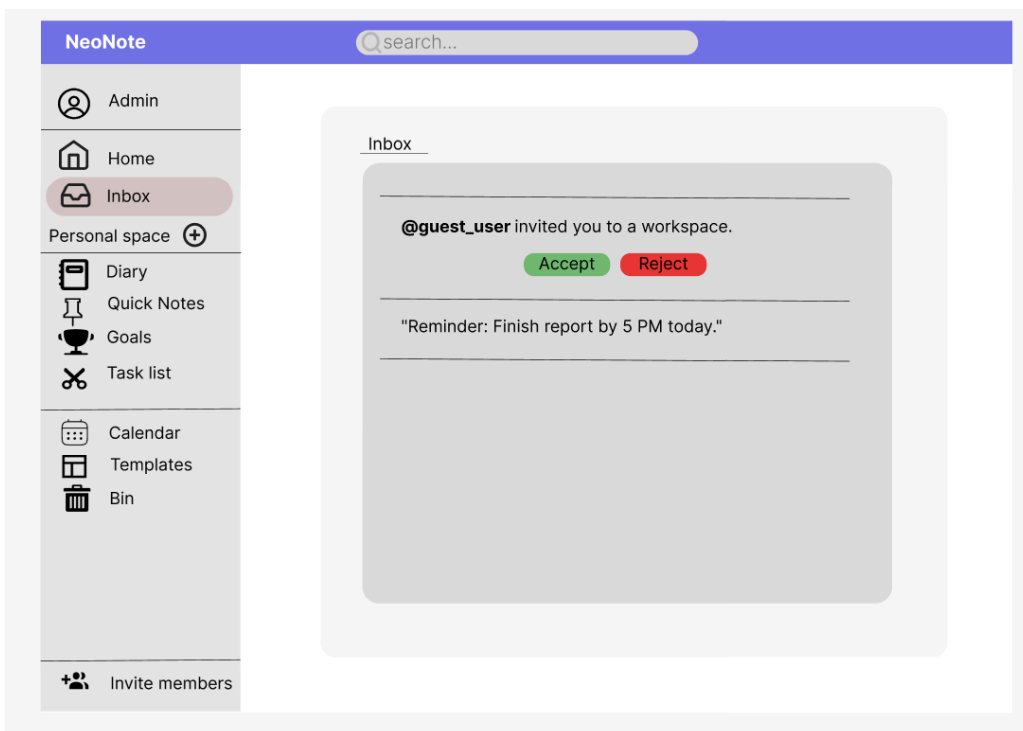


Fig 5.4 Inbox page

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